SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA13**

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:25**
Duration : 1 Hour

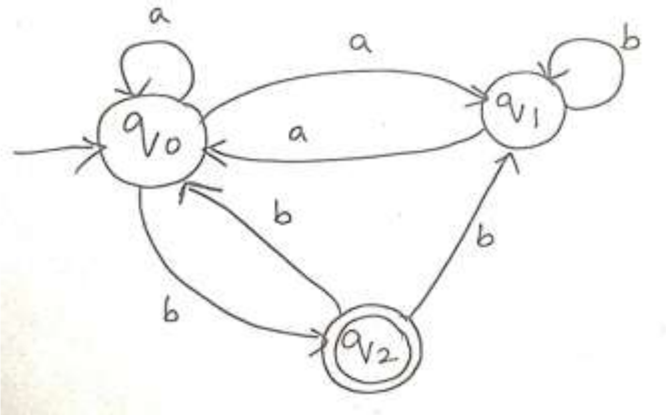
Analytical Problem

Code of Conduct:

I **Shaik Mohammed Waseem Rayan - 192373004 (***Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Shaik Mohammed Waseem Rayan

1.Consider the NFA given below which accepts some language L over the alphabet Σ={a,b}. Construct a DFA that represents the same language L



Construct the equivalent DFA for the given NFA

NFA

States:

- q_0 (Start)
- q_1
- q_2 (Final)

Alphabet:

$\Sigma = \{a,b\}$

Transitions:

State	a	b
q_0	q_0,q_1	q_2
q_1	q_0	q_1
q_2	-	q_0,q_1

Subset Construction

DFA States

- $A=\{q_0\}$
- $B=\{q_0,q_1\}$

- $C=\{q_2\}$
- $D=\{q_0,q_1,q_2\}$

Final states are those containing q_2.

DFA Transition Table

DFA State	a	b
$\rightarrow A=\{q_0\}$	B	C
$B=\{q_0,q_1\}$	B	D
$*C=\{q_2\}$	∅	B
$*D=\{q_0,q_1,q_2\}$	B	D
∅	∅	∅

Initial State = A
Final States = C,D

2.Let L be the language of words of length atleast 2 whose last two letters are difference from each other. For example abaab is in L but not bb, Construct an NFA for L, Then draw the equivalent DFA.

Language: Strings of length at least 2 whose last two symbols are different

Language
$L=\{w\in\{a,b\}^* \mid$ last two symbols are **ab** or **ba**$\}$
Examples
Accepted

- ab
- ba
- abab
- abaab

Rejected

- aa
- bb
- a
- b

NFA
States

- q_0 (Start)
- q_1
- q_2 (Final)

Transitions
q0 --a--> q0
q0 --b--> q0

q0 --a--> q1
q1 --b--> q2

q0 --b--> q1
q1 --a--> q2
q_2 is accepting.

Equivalent DFA
States represent last character.

State	a	b
→S	A	B
A	AA	AB*
B	BA*	BB
AA	AA	AB*
AB*	BA*	BB
BA*	AA	AB*
BB	BA*	BB

Final States
- AB
- BA

3.Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string 01011 is accepted by the NFA

NFA for strings ending with 00 or 11
Formal Definition
$M=(Q,\Sigma,\delta,q_0,F)$
Where
$Q=\{q_0,q_1,q_2\}$
$\Sigma=\{0,1\}$
Start=q_0
Final=$\{q_2\}$

Transition Function
$\delta(q0,0)=\{q0,q1\}$
$\delta(q0,1)=\{q0,q1\}$

δ(q1,0)={q2}
δ(q1,1)={q2}

δ(q2,0)=∅
δ(q2,1)=∅
Accept only when the two guessed last bits are equal.

Checking string 01011
q0
↓0
q0
↓1
q0
↓0
q0
↓1
q1
↓1
q2
Since computation ends in **q_2**, the string **01011 is accepted**.

4.Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet Σ={0,1}. Construct a DFA that represents the same language.

DFA for strings containing 001 as a substring

Language
All strings over {0,1} containing **001**.

States

- q_0 Start
- q_1 matched "0"
- q_2 matched "00"
- q_3 matched "001" (Final)

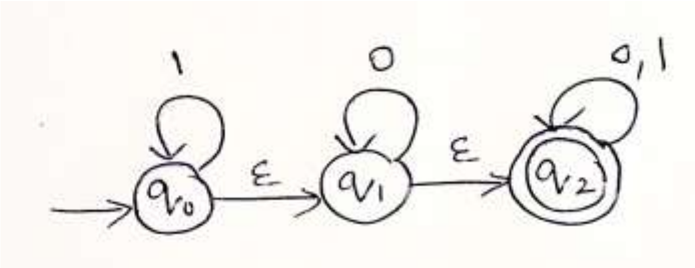
Transition Table

State	0	1
→q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
*q_3	q_3	q_3

Initial State
q_0
Final State

q_3

5.A NFA with ε-transitions is given below. Find ε-closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



5. ε-NFA to DFA

ε-Closures

Given ε-transitions

$q_0 \rightarrow q_1$

$q_1 \rightarrow q_2$

ε-closure(q_0)

{q0,q1,q2}

ε-closure(q_1)

{q1,q2}

ε-closure(q_2)

{q2}

DFA States

Let

A={q_0,q_1,q_2}

B={q_1,q_2}

C={q_2}

D=∅

Since q_2 is final,

Final states are A,B,C.

DFA Transition Table

DFA State	**0**	**1**
→*A={q_0,q_1,q_2}	C	A
*B={q_1,q_2}	C	C
*C={q_2}	C	C
D=∅	D	D

Initial State

A

Final States

A, B, C

SIMATS ENGINEERING
ASSESSMENT TOOL